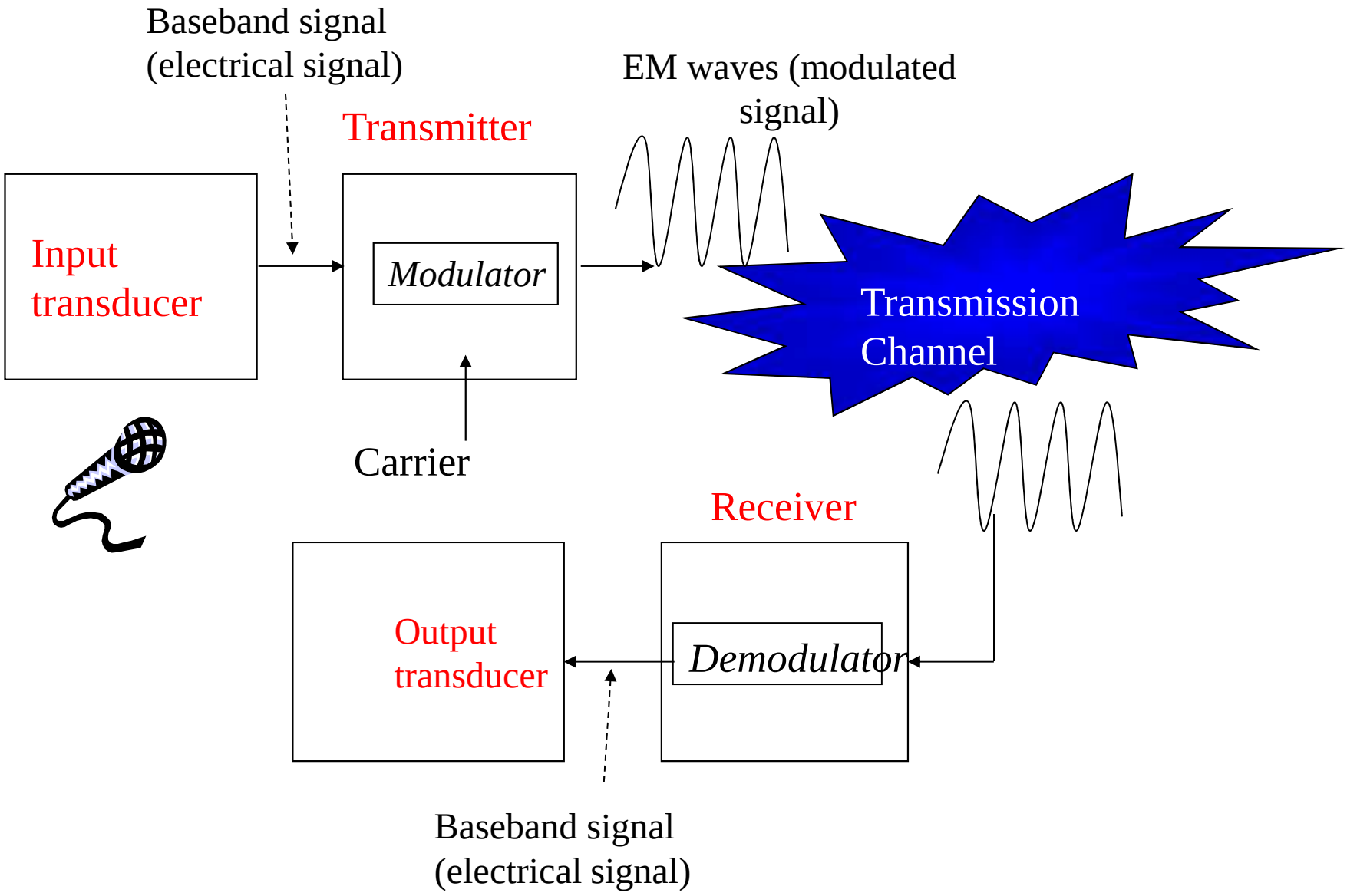
Selamat Pagi

Good Morning, How Are You today ?

Basic analog communications system



3. Modulasi Amplitudo



- **Modulasi amplitudo (AM = Amplitudo Modulation)** :
  Suatu teknik modulasi, dimana amplitudo sesaat gelombang sinyal pembawa (carrier), diubah-ubah sesuai dengan amplitudo sesaat gelombang sinyal informasi.

- Gelombang sinyal pembawa berfrekuensi tinggi, gelombang sinyal informasi berfrekuensi rendah.

- Amplitude Modulation is a process where the amplitude of a carrier signal is altered according to information in a message signal.

- The frequency of the carrier signal is usually much greater than the highest frequency of the input message signal.

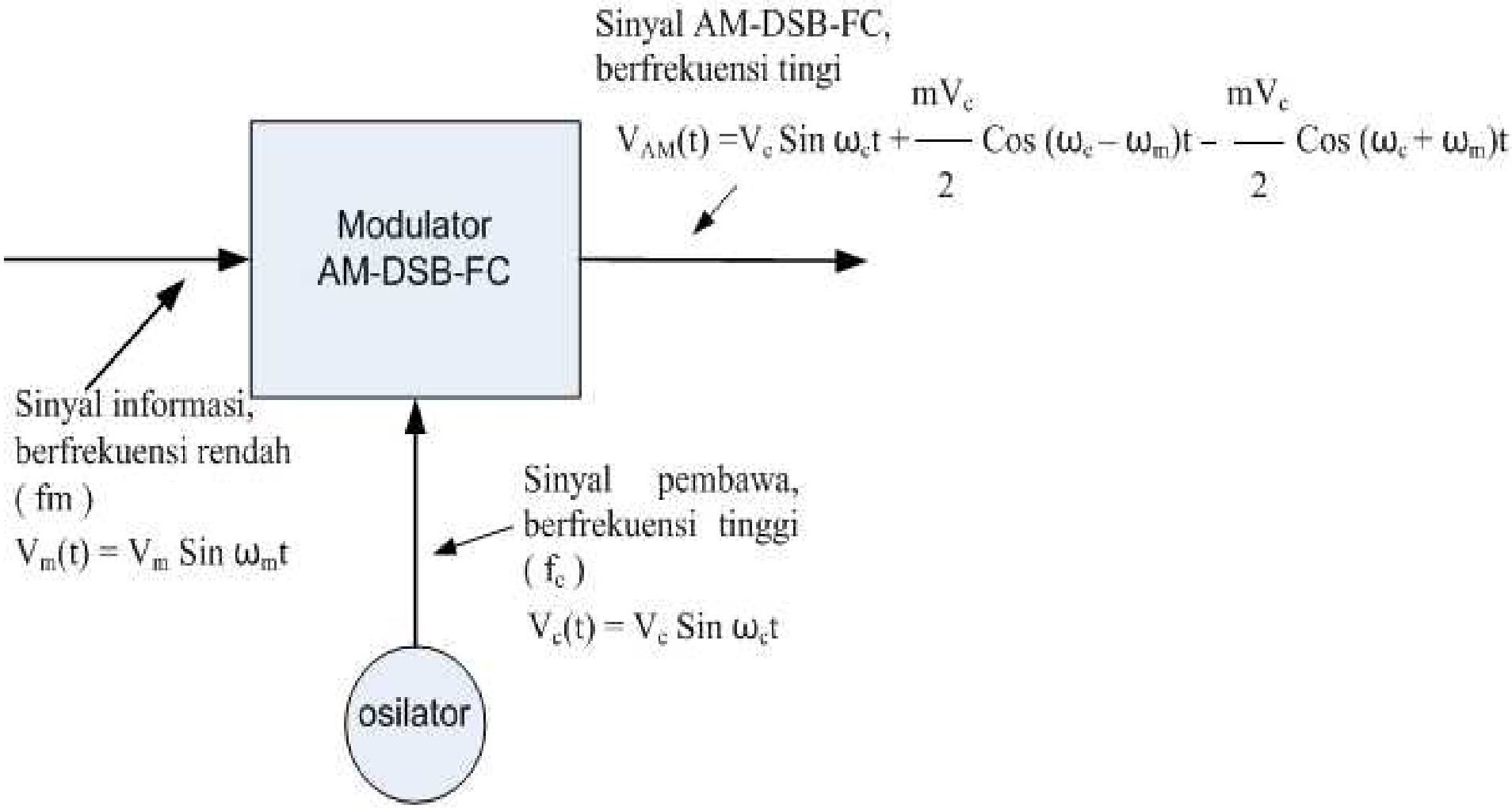
➢ Beberapa jenis /macam teknik modulasi amplitudo :

1) AM-DSB-FC: Amplitudo Modulation Double Side Band Full Carrier
2) AM-DSB-SC: Amplitudo Modulation Double Side Band Suppressed Carrier
3) AM-SSB-FC: Amplitudo Modulation Single Side Band Full Carrier
4) AM-SSB-SC: Amplitudo Modulation Single Side Band Suppressed Carrier
5) AM-VSB : Amplitudo Modulation Vestigial Side Band, yang digunakan sistem televisi untuk memodulasi sinyal gambar(video)

❖ AM-DSB-FC :

- ➢ Modulator AM merupakan perangkat nonlinier.
- ➢ Modulator memiliki dua input dan satu output.
- ➢ Input pertama sinyal gelombang informasi $V_m(t)$.
- ➢ Input kedua gelombang sinyal pembawa $V_c(t)$.
- ➢ Output gelombang sinyal termodulasi $V_{AM}(t)$.
- ➢ $V_m(t) = V_m \text{Sin}\omega_m t$.
- ➢ $V_c(t) = V_c \text{Sin}\omega_c t$.





Persamaan Gel AM-DSB-FC :

$$V_{AM}(t) = V_c \sin \omega_c t + \frac{mV_c}{2} \cos(\omega_c - \omega_m)t - \frac{mV_c}{2} \cos(\omega_c + \omega_m)t$$

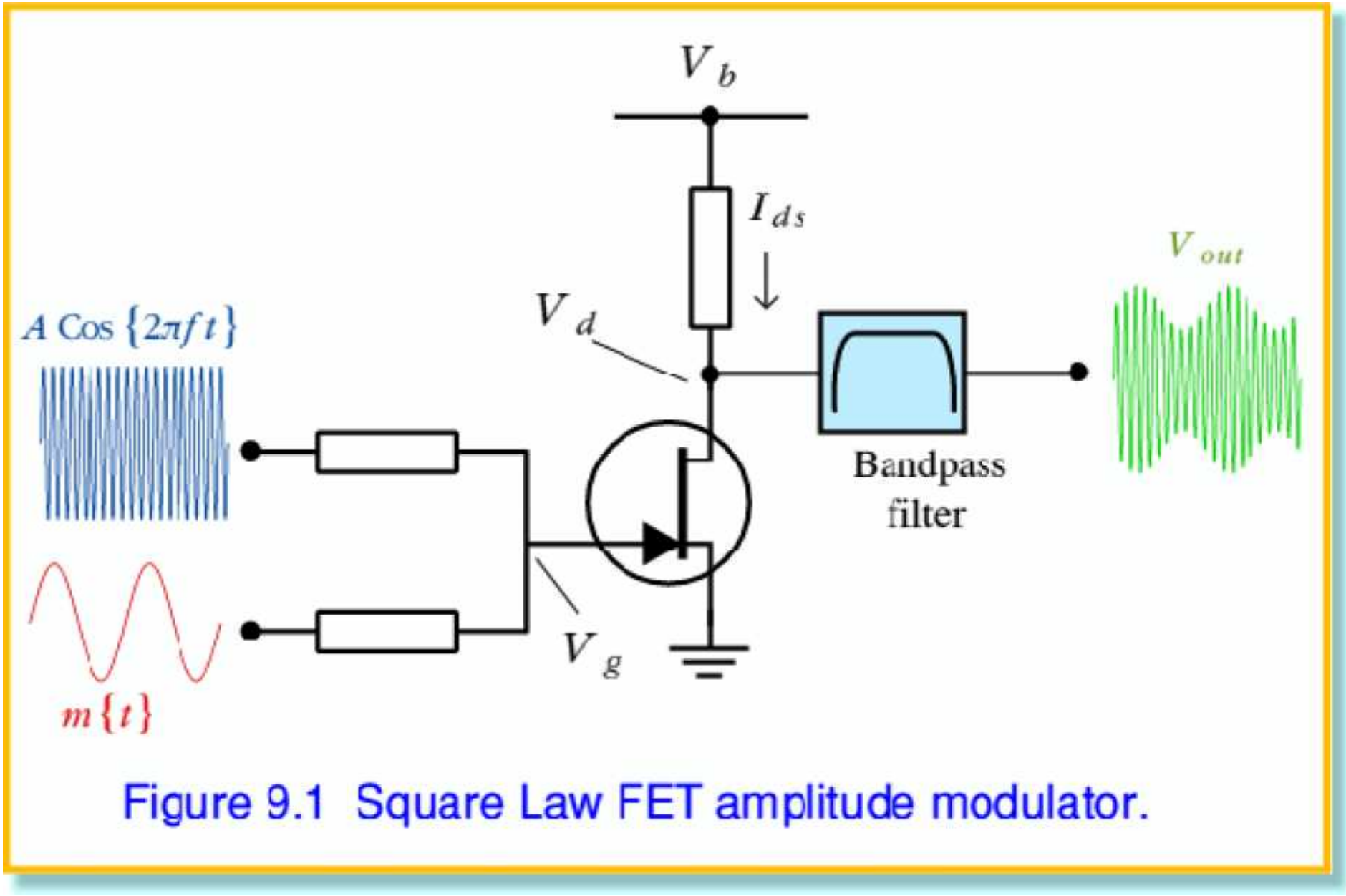
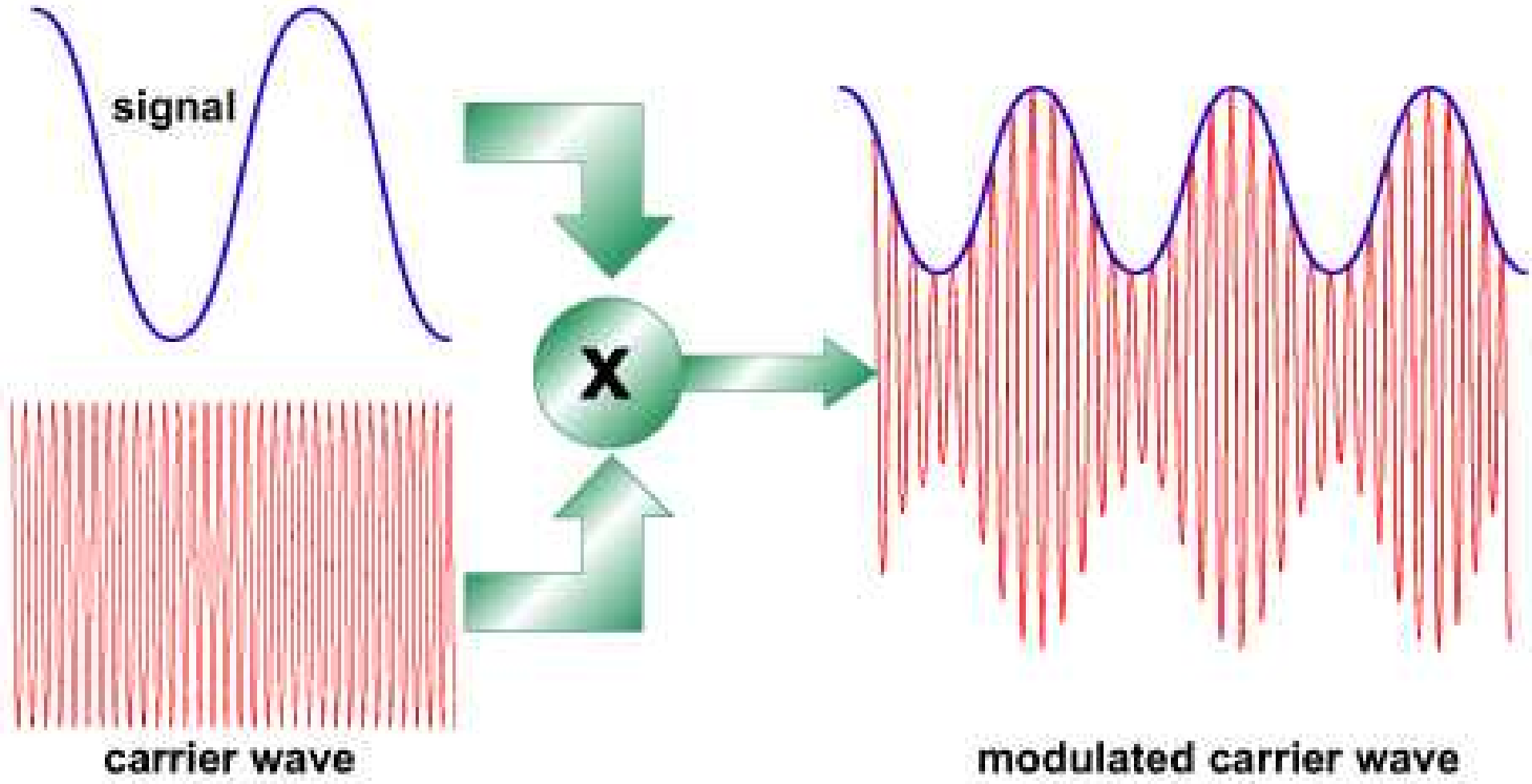
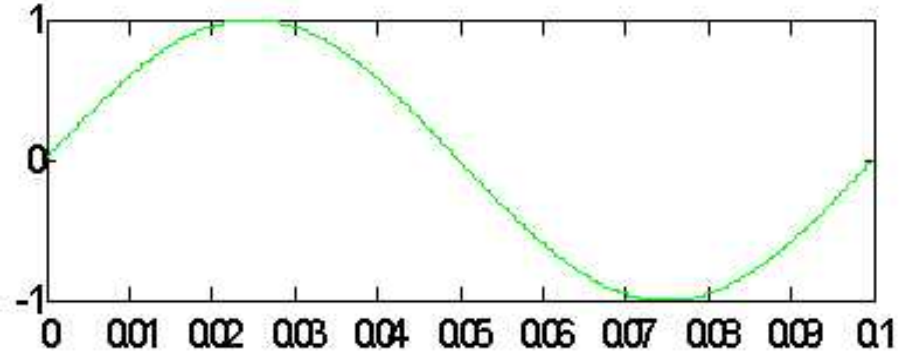


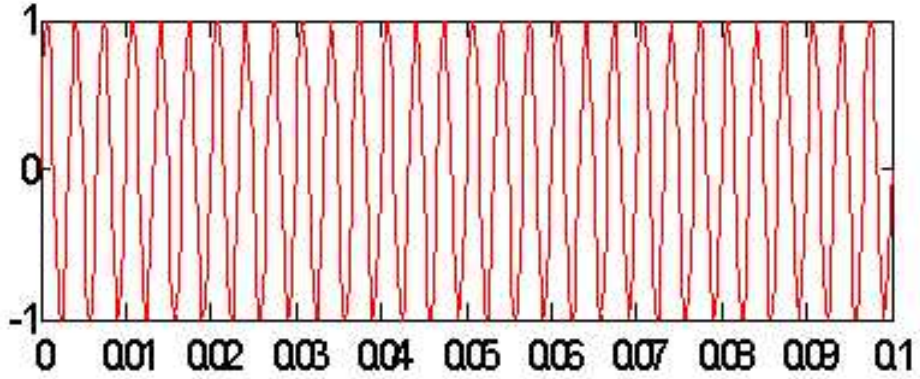
Figure 9.1 Square Law FET amplitude modulator.

Amplitude Modulation (AM)

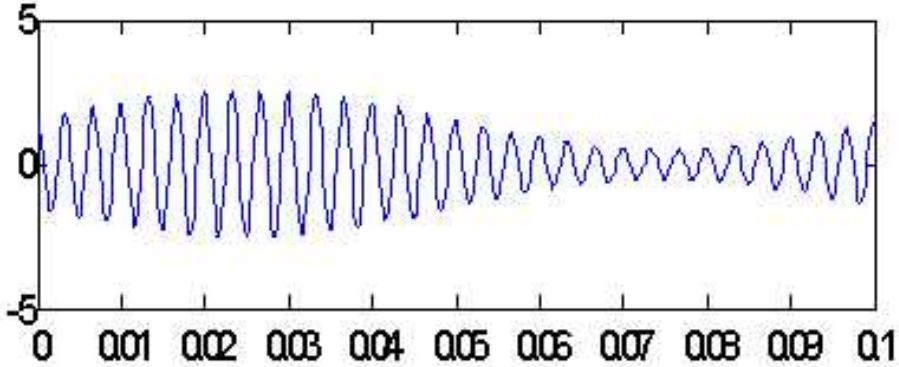




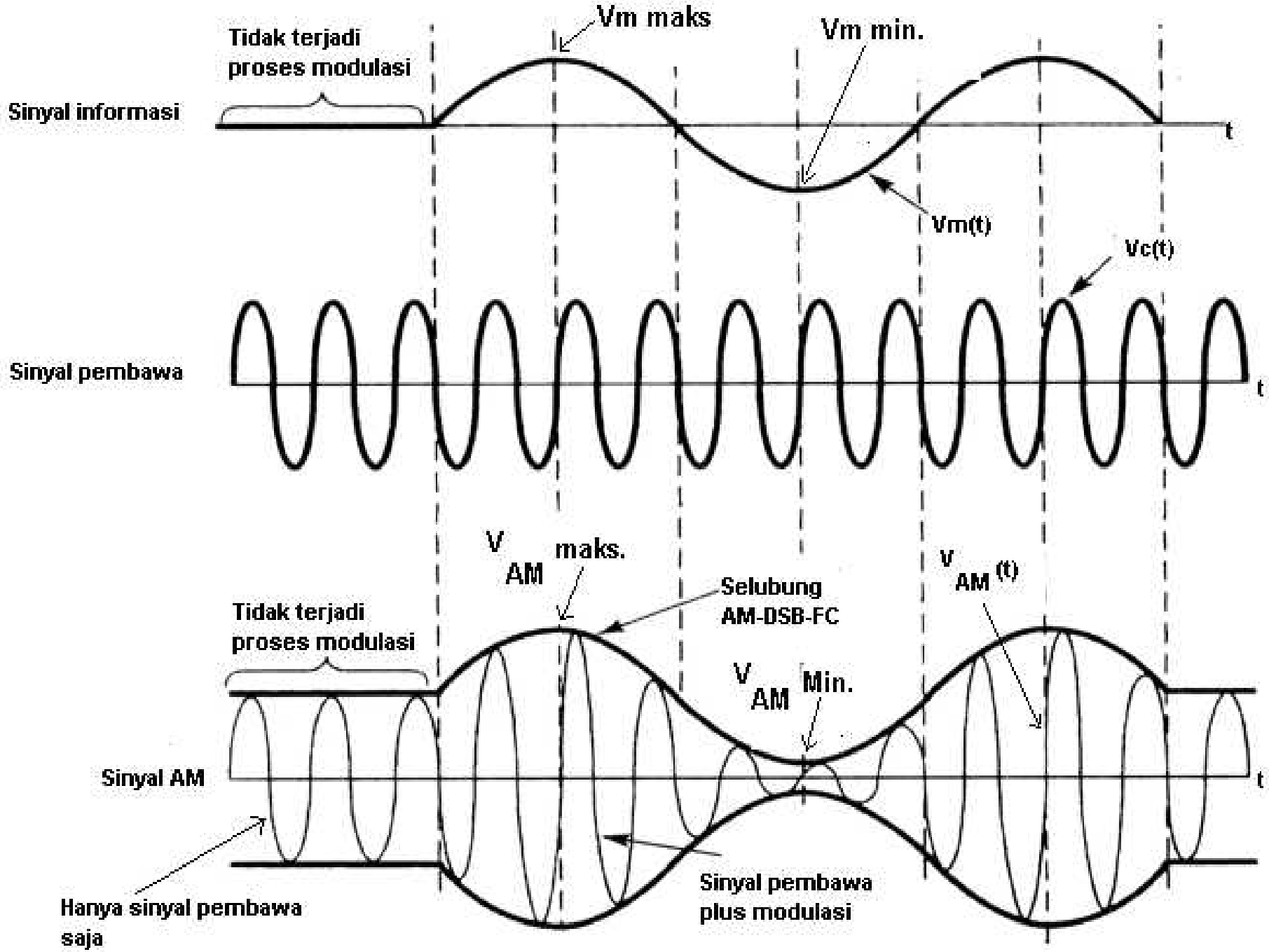
Gelombang Sinyal Informasi



Gelombang Sinyal Pembawa
(Carrier)



Gelombang Sinyal AM
(termodulasi)


Tidak terjadi
proses modulasi
Vm maks
Vm min.
Sinyal informasi
t
Vm(t)
Vc(t)
Sinyal pembawa
t
V AM maks.
Selubung
AM-DSB-FC
V AM (t)
Tidak terjadi
proses modulasi
V AM Min.
Sinyal AM
t
Hanya sinyal pembawa
saja
Sinyal pembawa
plus modulasi

➢ Persamaan Matematis Gelombang AM-DSB-FC

For $V_c(t) = V_c \cos 2\pi f_c t$

$$V_m(t) = V_m \cos 2\pi f_m t$$

Modulated Signal : $V_{AM}(t) = (V_c + V_m \cos 2\pi f_m t)(\cos 2\pi f_c t)$

$$V_{AM}(t) = V_c(1 + m \cos 2\pi f_m t)(\cos 2\pi f_c t)$$

Given : $(\cos X)(\cos Y) = \frac{1}{2}\cos(X - Y) + \frac{1}{2}\cos(X + Y)$

We get : $V_{AM}(t) = V_c \cos 2\pi f_c t + V_c m \cos 2\pi f_c t \cos 2\pi f_m t$

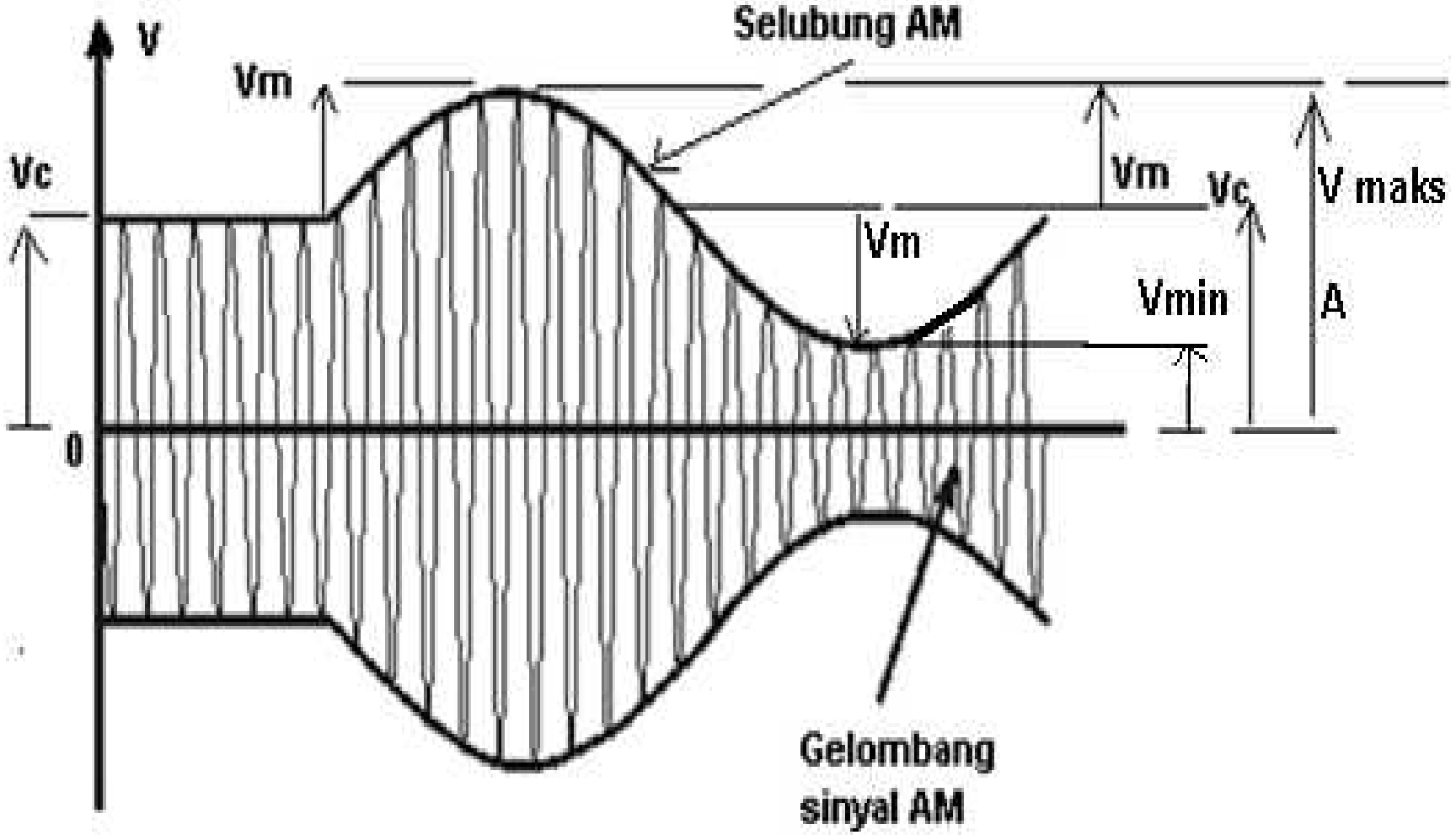
Which gives :

$$V_{AM}(t) = V_c \cos 2\pi f_c t + \frac{mVc}{2}[\cos 2\pi (f_c - f_m)t + \cos 2\pi (f_c + f_m)t]$$

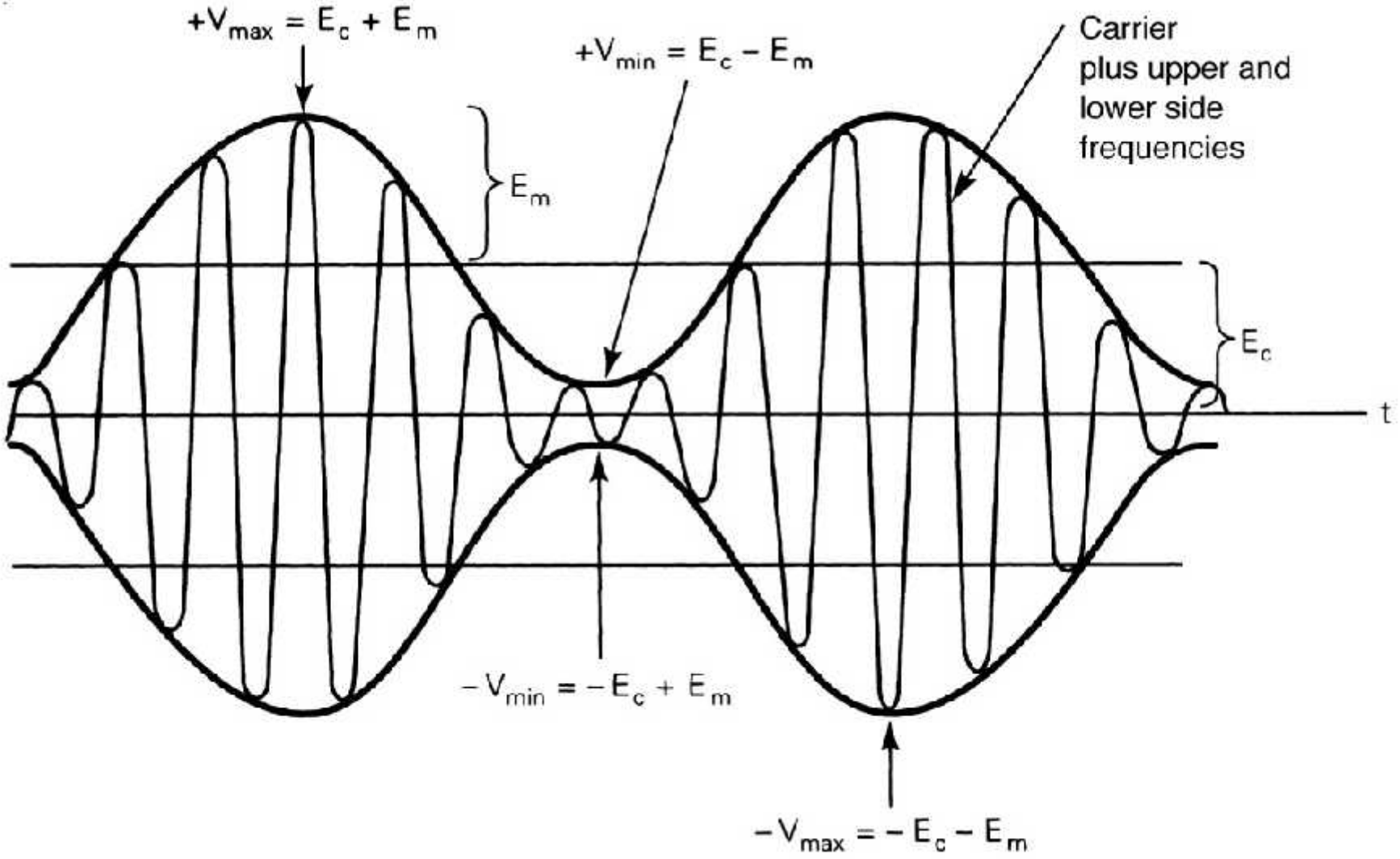
Dimana m = Indeks modulasi, $m = \frac{V_m}{V_c}$

➢ Persamaan Matematis Gelombang AM-DSB-FC

$$V_{AM}(t) = V_c \sin \omega_c t + \frac{mV_c}{2} \cos(\omega_c - \omega_m)t - \frac{mV_c}{2} \cos(\omega_c + \omega_m)t$$



Gelombang AM-DSB-FC



AMPLITUDE MODULATION (DSB_FC)

$$V_{max} = E_c + E_m \; ; \quad V_{min} = E_c - E_m$$

$$E_m = \frac{1}{2}(V_{max} - V_{min})$$

$$E_c = \frac{1}{2}(V_{max} + V_{min})$$

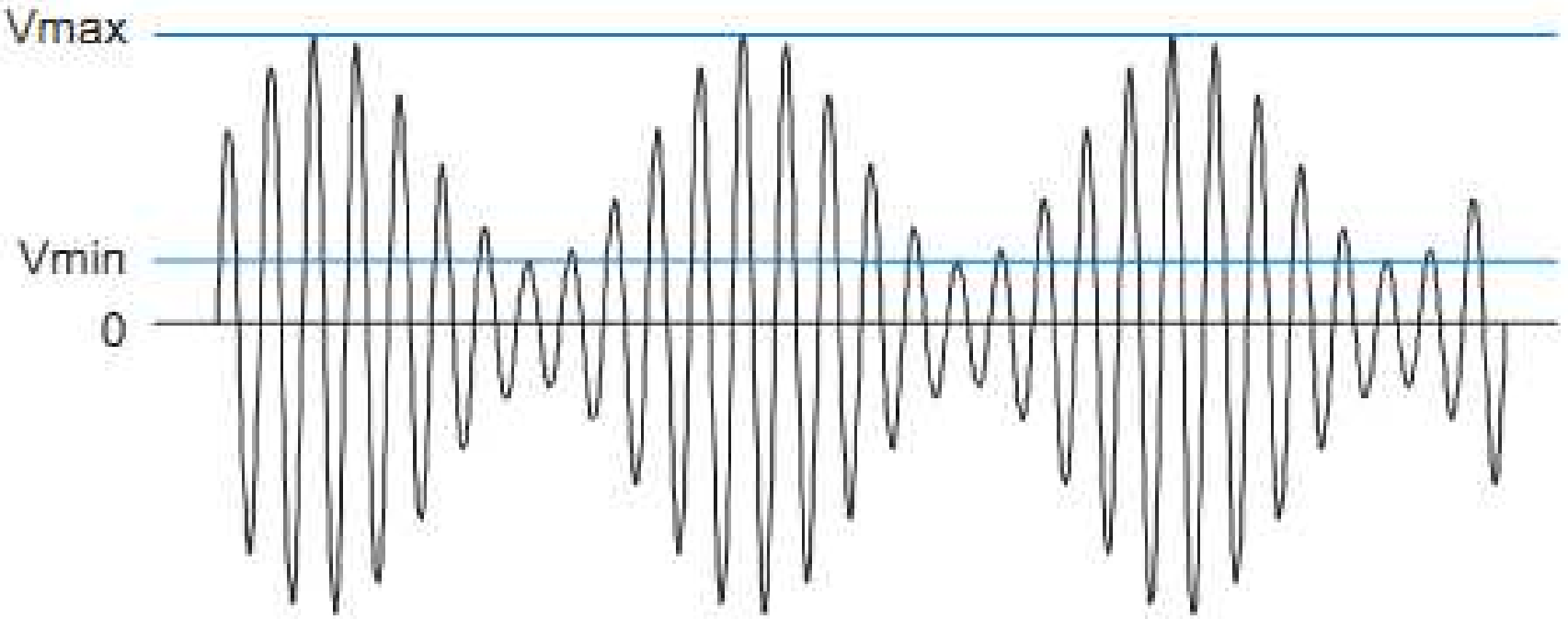
$$m = \frac{E_m}{E_c} = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$$

$$E_{usf} = E_{lsf} = \frac{E_m}{2} = \frac{1}{4}(V_{max} - V_{min})$$

E_{USF} = PEAK AMPLITUDE OF THE UPPER SIDE FREQUENCY

E_{LSF} = PEAK AMPLITUDE OF THE LOWER SIDE FREQUENCY

Measuring modulation index



$$m = \frac{\text{Vmax} - \text{Vmin}}{\text{Vmax} + \text{Vmin}}$$

$$m = \frac{V_m}{V_c}$$

$$\% m = \frac{V_m}{V_c} \text{ X 100 \%}$$

$$\% \mathrm{m} = \frac{\mathrm{Vmax} - \mathrm{Vmin}}{\mathrm{Vmax} + \mathrm{Vmin}} \text{ X 100 \%}$$

- Untuk Modulasi AM, indeks modulasi maksimum sebesar 1 atau 100 %, jika indeks modulasi lebih besar dari 1 atau lebih besar dari 100 %, maka akan terjadi ***over modulasi*****.**

- **Definition:** The **percentage of positive modulation** on an AM signal is

$$\%\ \text{Positive Modulation} = \frac{A_{\max} - A_c}{A_c} \times 100 = \max\left[m(t)\right] \times 100$$

- The **percentage of negative modulation** on an AM signal is

$$\frac{A_c - A_{\min}}{A_c} \times 100 = -\min\left[m(t)\right] \times 100$$

- The **percentage of overall modulation** is

$$\%\ \text{Modulation} = \frac{A_{\max} - A_{\min}}{2A_c} \times 100 = \frac{\max\left[m(t)\right] - \min\left[m(t)\right]}{2} \times 100$$

$A_{\max}$ - Maximum value of $A_c[1 + m(t)]$

$A_{\min}$ - Minimum value of $A_c[1 + m(t)]$

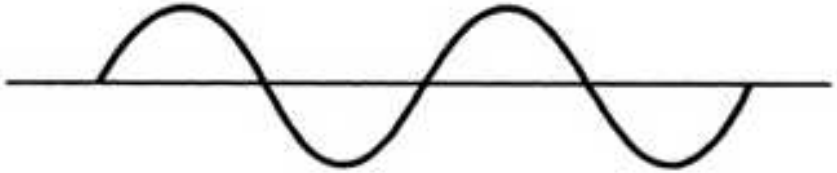
A_c - Level of AM envelope in the absence of modulation [i.e., $m(t) = 0$]

If $m(t)$ has a peak positive values of +1 and a peak negative value of -1

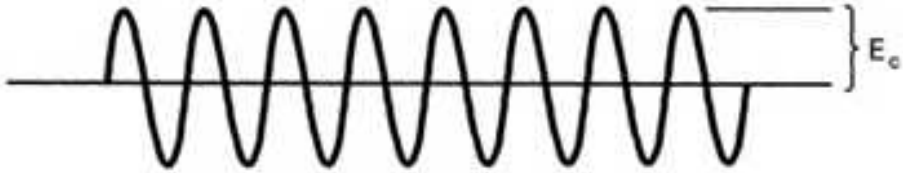
AM signal → 100% modulated

AM – Percentage Modulation

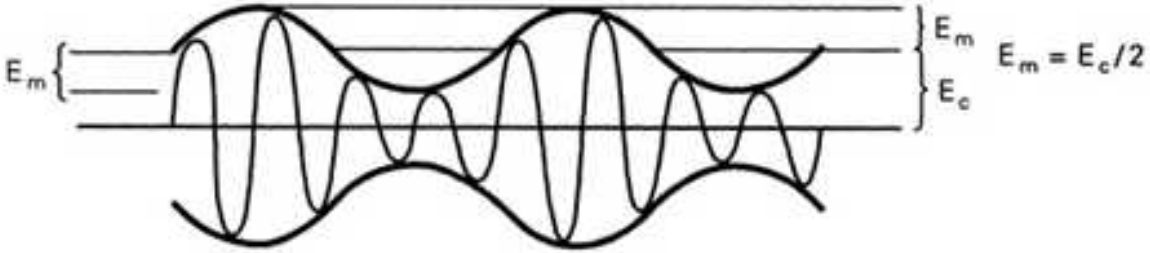
Modulating Signal (a)



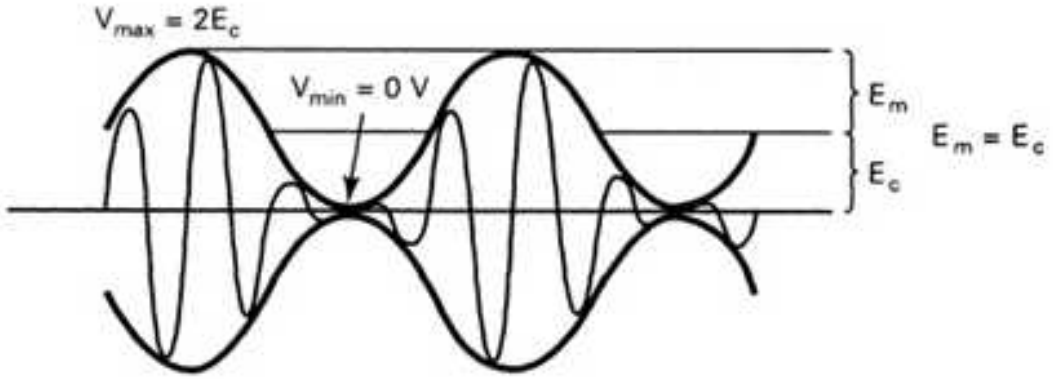
Unmodulated Carrier (b)



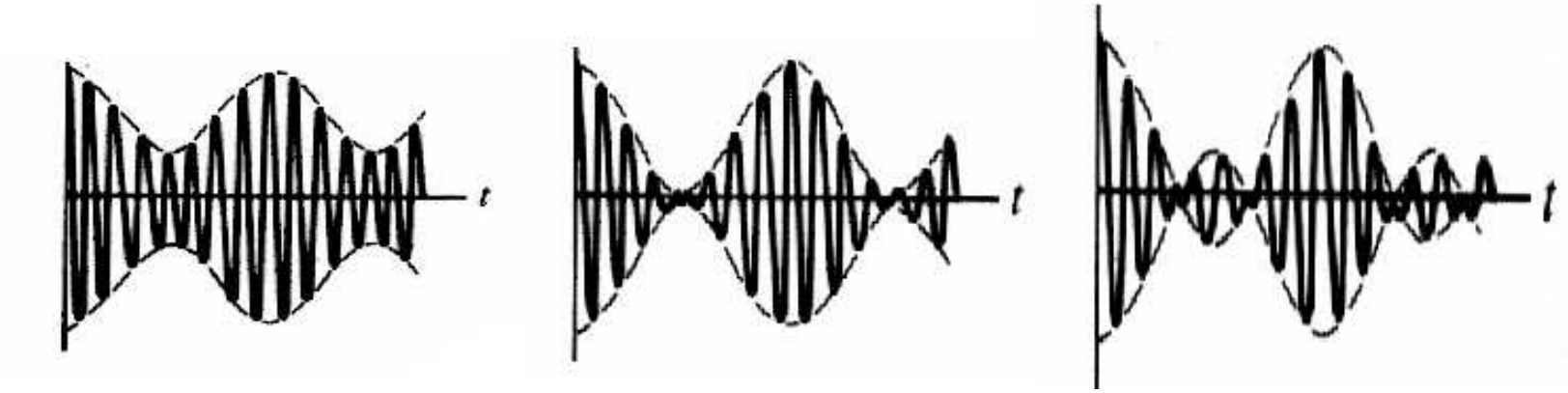
50% Modulation (c)



100% Modulation (d)



AM – Percentage Modulation



Under modulated (<100%)

100% modulated

Over Modulated (>100%)

Envelope Detector

Can be used

Envelope Detector

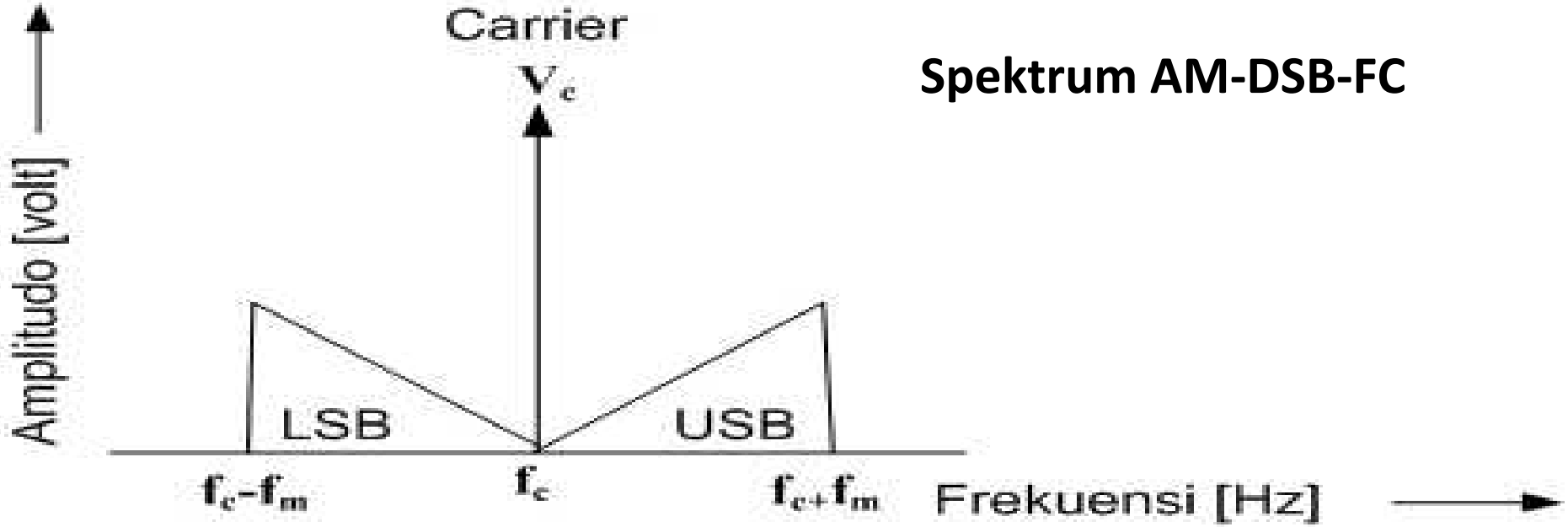
Gives Distorted signal

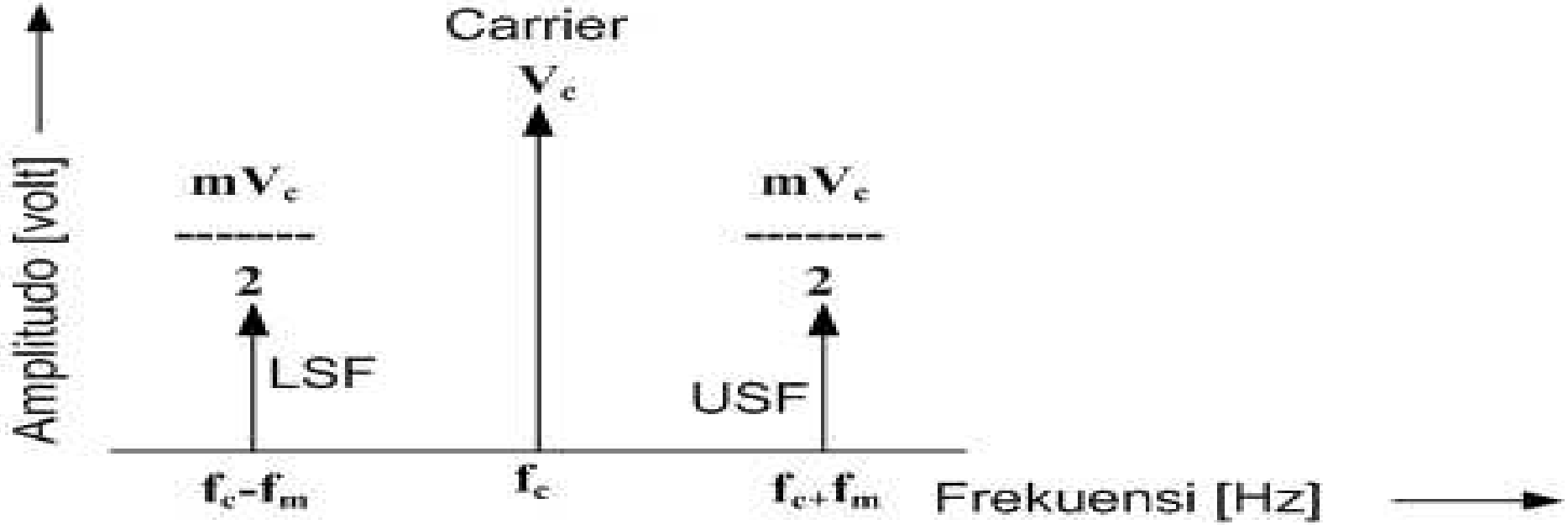
- Spektrum, Bandwidth dan Daya AM-DSB-FC

$$V_{AM}(t) = V_c \sin \omega_c t + \frac{mV_c}{2} \cos(\omega_c - \omega_m)t - \frac{mV_c}{2} \cos(\omega_c + \omega_m)t$$

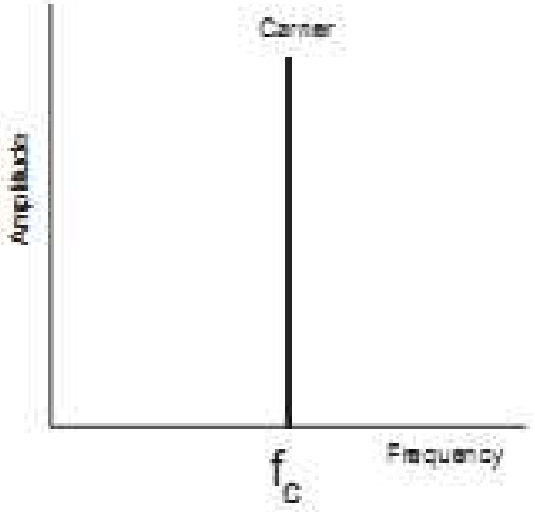
- Terlihat bahwa sinyal tersebut mengandung komponen frekuensi :
  - Frekuensi pembawa, ω_c, ($\omega = 2\pi f$), 2π = konstan.
  - Frekuensi-frekuensi sisi atas USF (Upper Side Frequencies) atau USB (Upper Side Band), $\omega_c + \omega_m$.
  - Frekuensi-frekuensi sisi bawah LSF (Lower Side Frequencies) atau LSB (Lower Side Band), $\omega_c - \omega_m$

Spektrum AM-DSB-FC

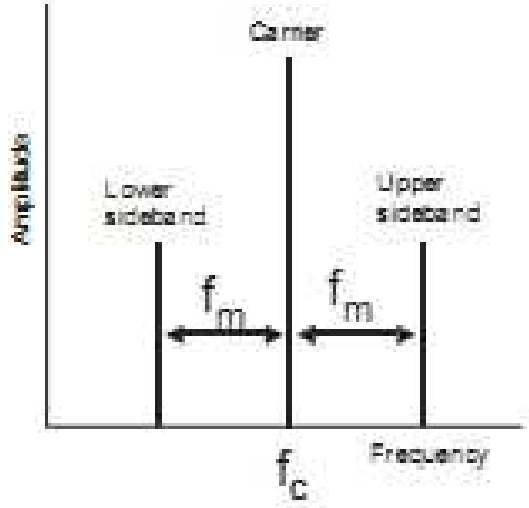




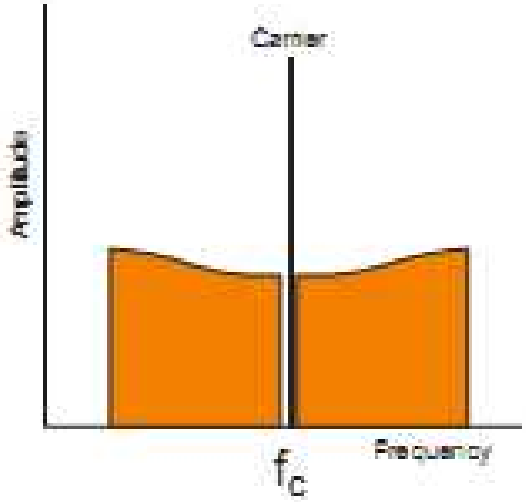
AM spectrum



unmodulated carrier



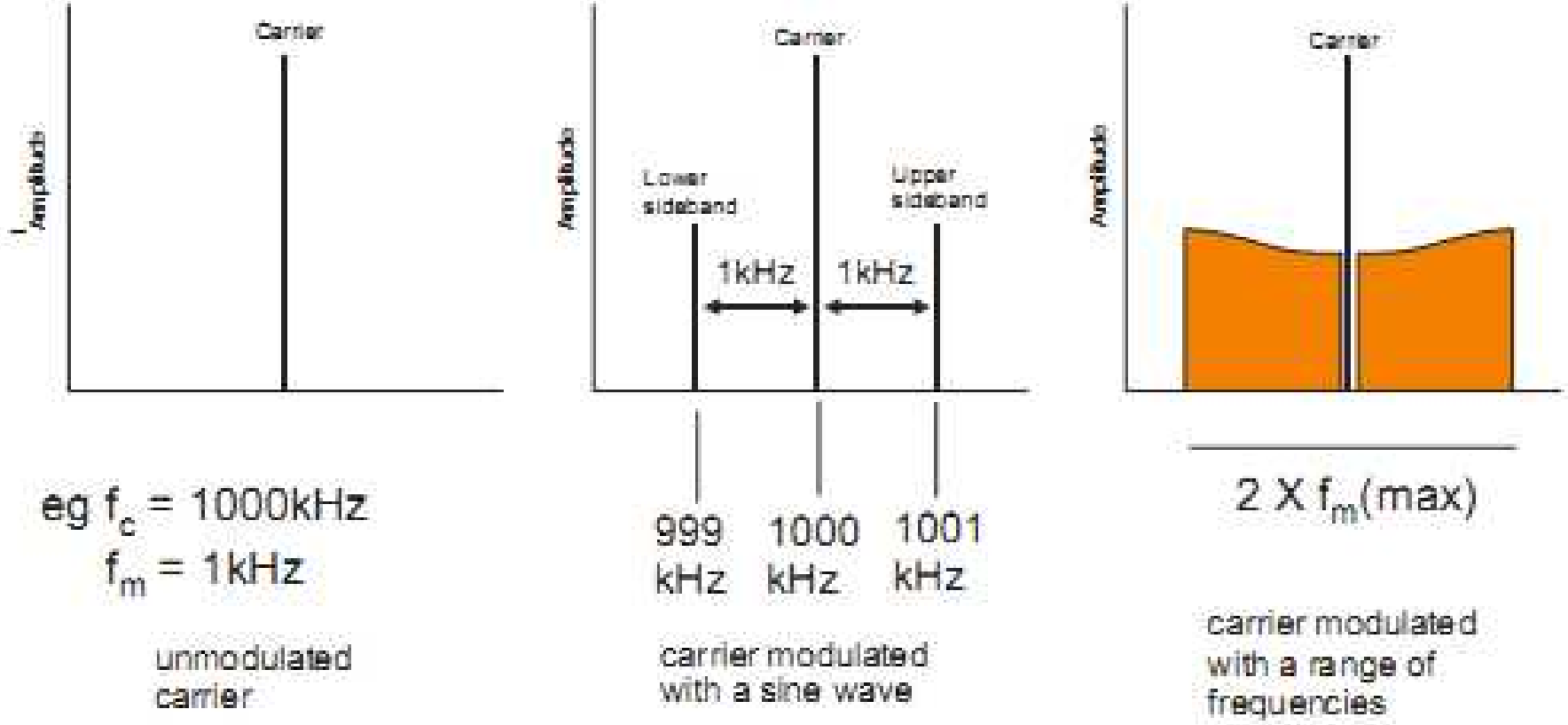
carrier modulated with a sine wave

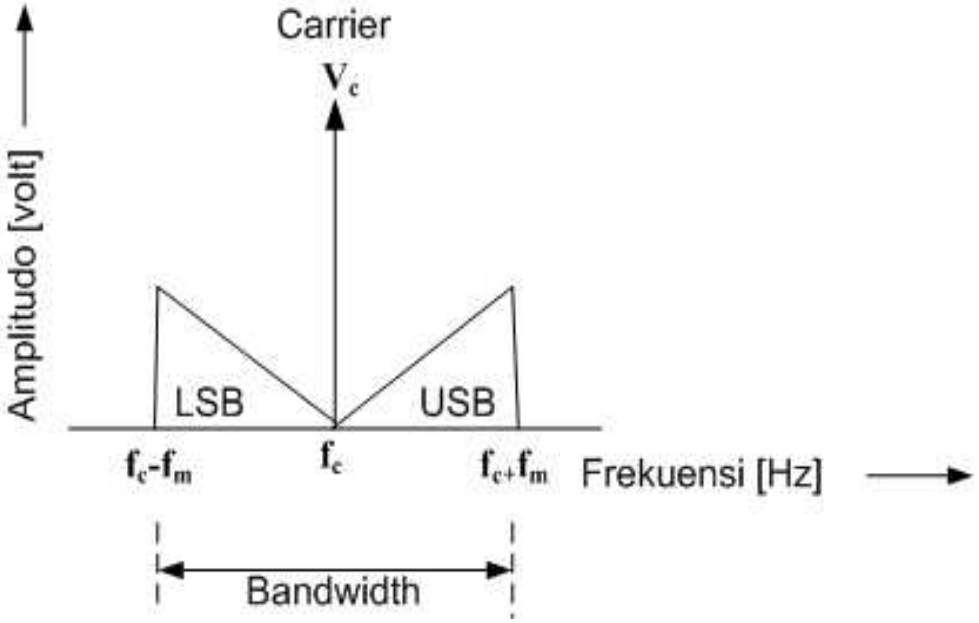


carrier modulated with a range of frequencies

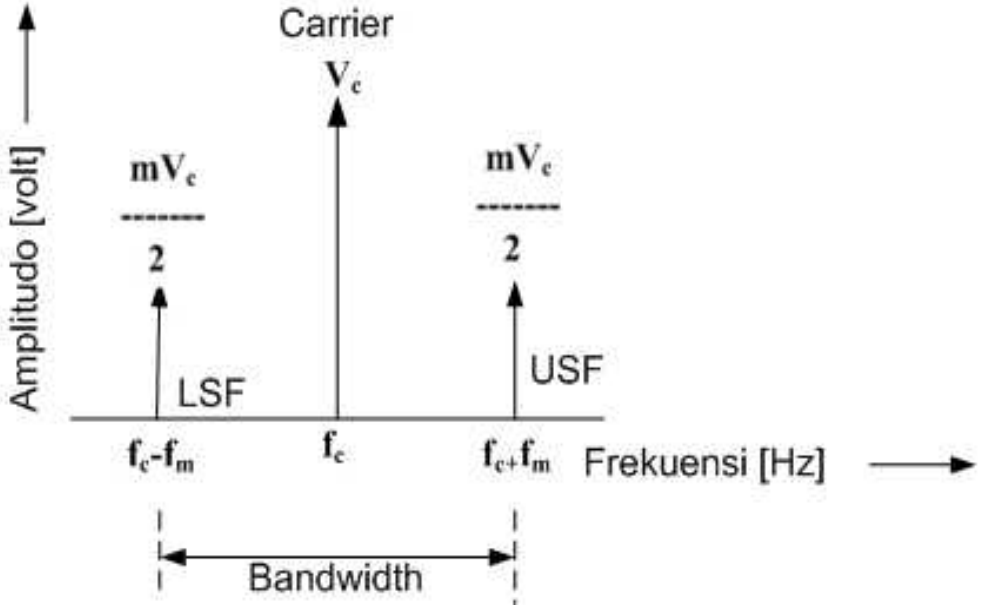
AM spectrum

- Example: f_c = 1000 kHz, f_m = 1 kHz





Banwidth,
Bw = 2 fm



Daya pancar total AM-DSB-FC :

$$P_t = P_c\left(1+\frac{m^2}{2}\right)$$

DSB-FC EXAMPLE

1. FOR AN AM DSB-FC MODULATOR, WITH CARRIER FREQUENCY OF 100 kHz, AND A MAXIMUM MODULATING SIGNAL OF 5 kHz (artinya sinyal pemodulasi sebesar dari 0 s/d 5 KHz)

DETERMINE:

a) FREQUENCY LIMITS FOR THE UPPER AND LOWER SIDEBANDS

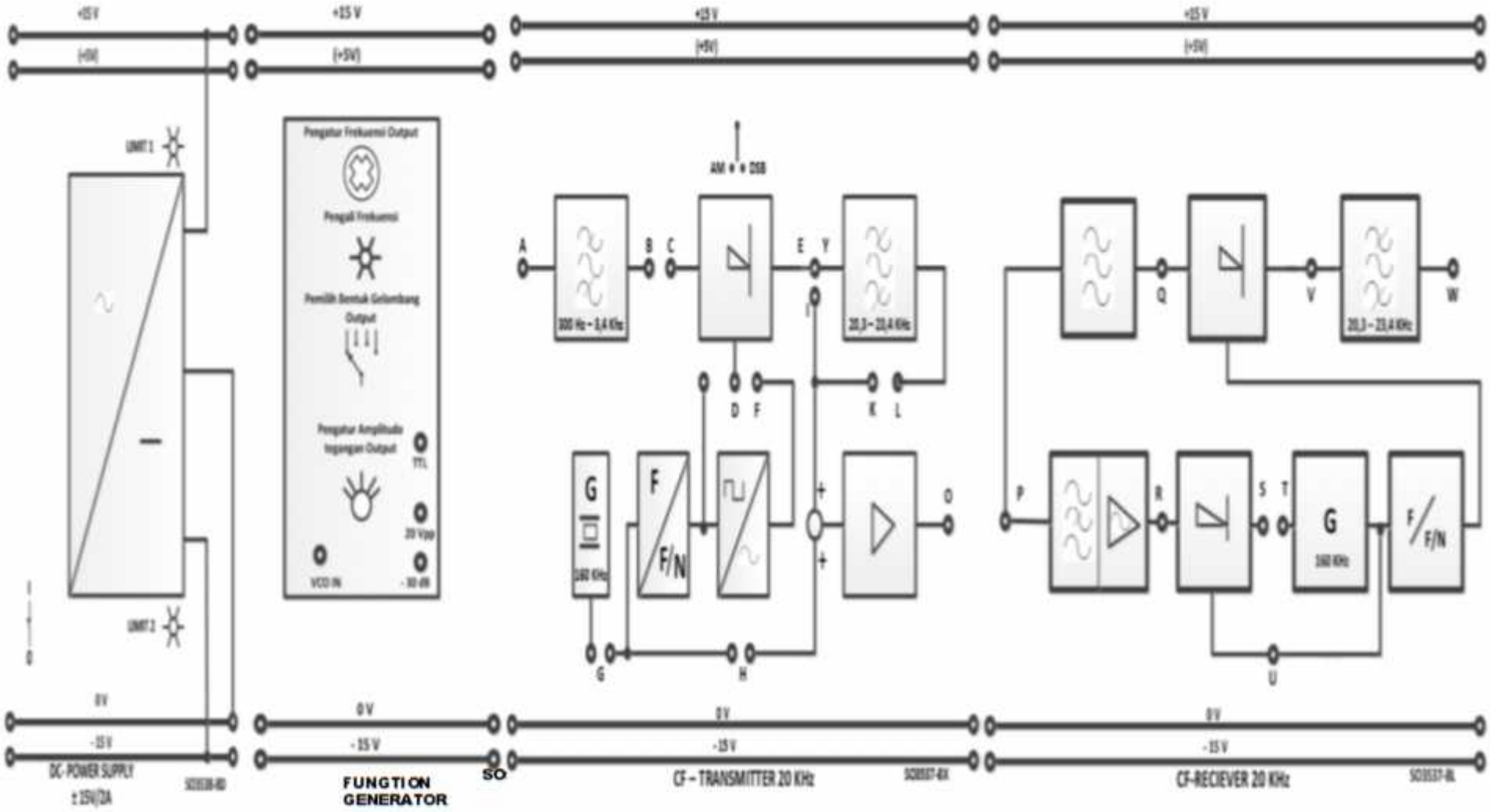
Jawab : (100 - 5) kHz TO 100 kHz = 95 kHz TO 100 kHz = LSB
100 kHz TO (100 + 5) kHz = 100 kHz TO 105 kHz = USB

b) BANDWITH OF THE MODULATED SIGNAL

Jawab : B = 2 fm = 2 x 5 kHz = 10 kHz

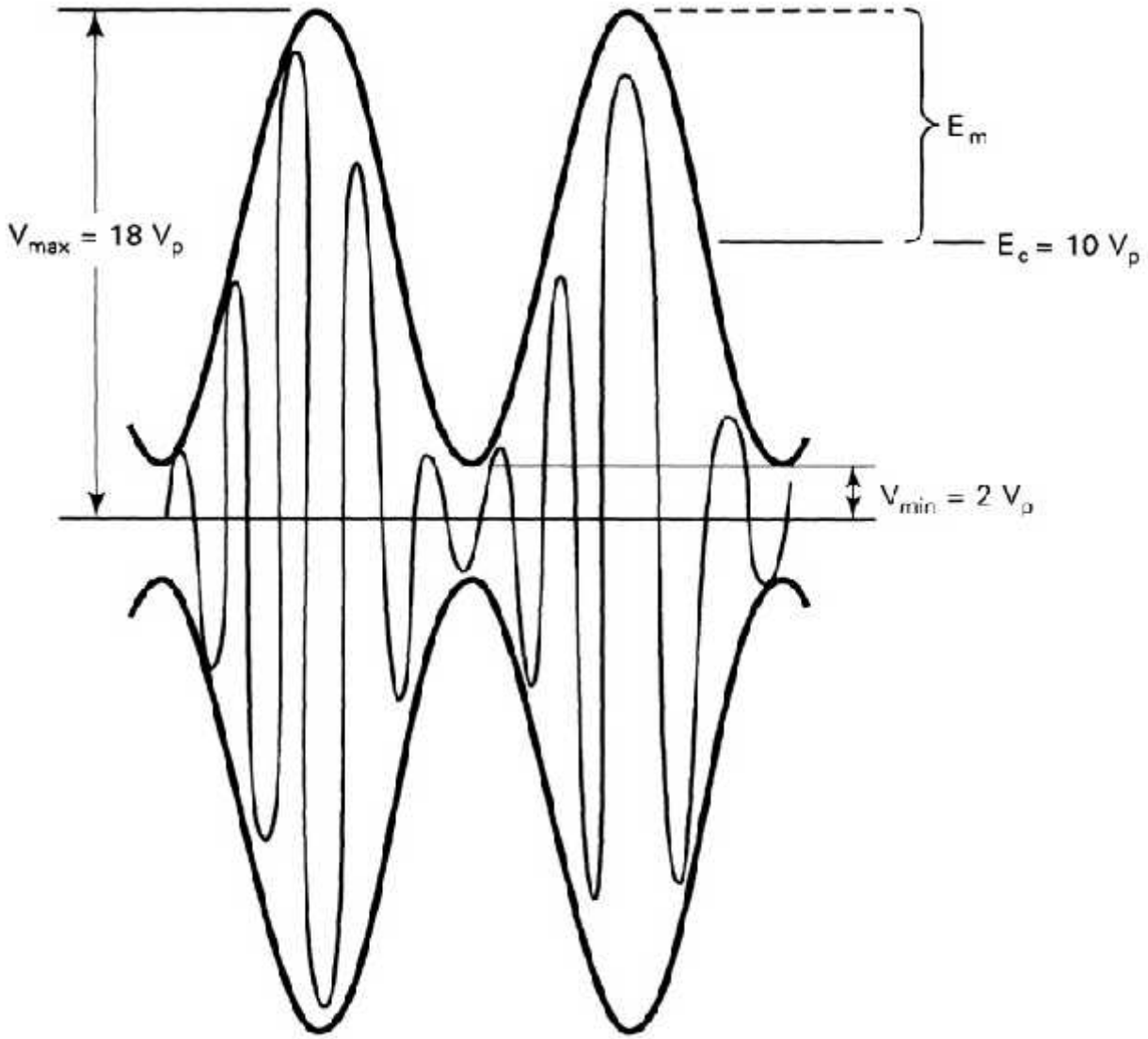
c) UPPER AND LOWER SIDE FREQUENCIES WHEN MODULATING SIGNAL IS A 3 kHz TONE (artinya sinyal pemodulasi berfrekuensi tunggal).

Jawab : (100 - 3) kHz = 97 kHz = LSF
(100 + 3) kHz = 103 kHz = USF



Gambar 1.10 Transmitter & Reciever AM

2. FOR THE AM WAVEFORM BELOW:



DETERMINE:

a) PEAK AMPLITUDE OF THE UPPER AND LOWER SIDE FREQUENCIES

Jawab :

$$V_{usf} = V_{lsf} = \frac{V_m}{2} = \frac{1}{4}(V_{\max} - V_{\min})$$

$$V_{usf} = V_{lsf} = \frac{1}{4}(18 - 2) = 4Volt$$

b) PEAK AMPLITUDE OF THE UNMODULATED CARRIER

Jawab :

$$V_c = \frac{1}{2}(V_{\max} + V_{\min}) = \frac{1}{2}(18 + 2) = 10\,Volt$$

c) PEAK CHANGE IN THE AMPLITUDE OF THE ENVELOPE

Jawab :

$$V_m = \frac{1}{2}(V_{\max} - V_{\min}) = \frac{1}{2}(18 - 2) = 8Volt$$

d) COEFFICIENT INDEX

Jawab :

$$= \frac{V_m}{V_c} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{8}{10} = 0.8$$

e) PERCENT MODULATION

Jawab :

$$m = \frac{V_m}{V_c} x\ 100\ \% = 0.8 x 100\ \% = 80\ \%$$

$$M = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} x100\ \% = \frac{18 - 2}{18 + 2} x100\ \% = 80\ \%$$

3. ONE INPUT TO A CONVENTIONAL MODULATOR IS A 500 kHz CARRIER WITH AN AMPLITUDE OF 20 Vp. THE SECOND INPUT IS A 10 kHz MODULATING SIGNAL THAT IS OF SUFFICIENT AMPLITUDE TO CAUSE A CHANGE IN THE OUTPUT WAVE OF $\pm\ 7.5\ Vp$.

DETERMINE:

a) UPPER AND LOWER SIDE FREQUENCIES

Jawab :

(500 + 10) kHz = 510 kHz = USF
(500 - 10) kHz = 490 kHz = LSF

b) MODULATION COEFFICIENT AND PERCENT MODULATION

Jawab :

$$m = \frac{7.5}{20} = 0.375$$

$$m = \frac{7.5}{20} x100\ \% = 37.5\%$$

C) PEAK AMPLITUDE OF THE MODULATED CARRIER

Jawab :

Vc (MODULATED) = Vc (UNMODULATED) = 20 Vp

d) UPPER AND LOWER SIDE FREQUENCY VOLTAGES

Jawab :

$$V_{usf} = V_{lsf} = \frac{V_m}{2} = \frac{mV_c}{2} = \frac{0.375\ (20)}{2} = 3.75\,Vp$$

e) MAXIMUM AND MINIMUM AMPLITUDES OF THE ENVELOPE

Jawab :

$$V_{max} = V_c + V_m \;; \quad V_{min} = V_c - V_m$$

$$V_{max} = 20 + 7.5 = 27.5Vp$$

$$V_{min} = 20 - 7.5 = 12.5Vp$$

4. FOR AN AM DSB-FC WAVE WITH A PEAK UNMODULATED CARRIER VOLTAGE Vc = 10 Vp, A LOAD RESISTANCE OF RL = 10 Ohms, AND A MODULATION INDEX OF 1,

DETERMINE:

a) CARRIER POWER

Jawab :

$$P_c = \frac{V_c{}^2}{R} = \frac{10^{\ 2}}{(10\ )} = 10\,W$$

b) UPPER AND LOWER SIDEBAND POWER

Jawab :

$$P_{usb} = P_{lsb} = \frac{m^{\ 2}}{4} P_c = \frac{1(10\ )}{4} = 2,5\,W$$

c) TOTAL SIDEBAND POWER

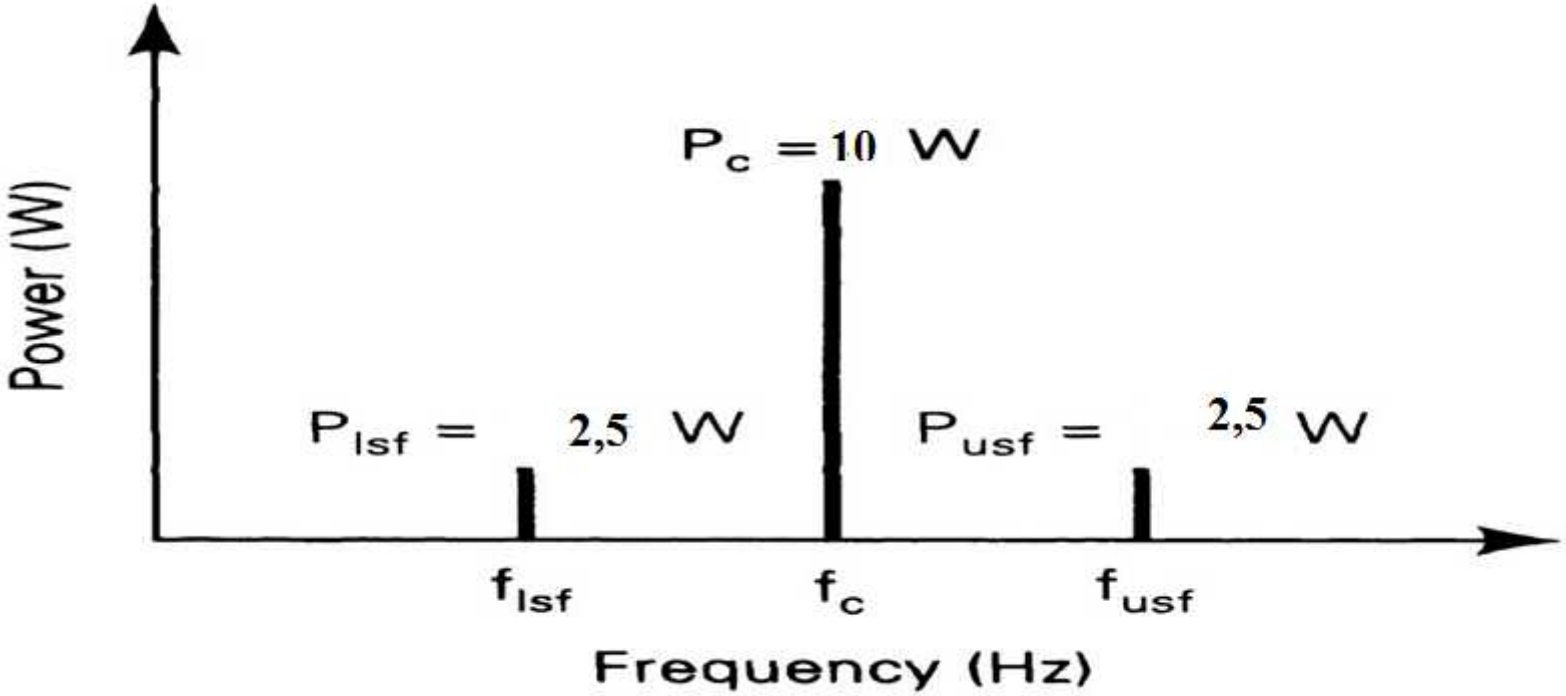
$$P_{usb} + P_{lsb} = \frac{m^{\ 2}}{2} P_c = \frac{1(10\ )}{2} = 5\ W$$

d) TOTAL POWER IN THE MODULATED WAVE

Jawab :

$$P_t = P_c\left(1 + \frac{m^2}{2}\right) = 10\left(1 + \frac{1^2}{2}\right) = 15\ W$$

POWER SPECTRUM:



Thankyou, see you

next week,

ok

Bye-bye